

Theory of Computation

BSc Computer Science - Durham University

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October 2026

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1 Module Introduction

Theory of Computation has the module code COMP2181 and is led by Dr Barnaby Martin.

1.1 Assessment

This module is assessed through two summative benchtests, and one final exams, these can be seen in Table 1.

Title	Opening	Due	Feedback	Professor	Weight
Benchtest 1	Week 11	n/a	09/02/2027	Dr Dantchev and Dr Mertzios	17%
Benchtest 2	Week 20	n/a	07/05/2027	Dr Dantchev and Dr Mertzios	17%
Final Exam	May Exam	n/a	n/a	Dr Barnaby Martin	66%

Table 1: Assessments for Theory of Computation.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Tuesday	10:00 - 11:00	Lecture	CLC202	Dr Barnaby Martin
Wednesday	09:00 - 11:00	Problem Class	TLC116	n/a
Thursday	14:00 - 15:00	Lecture	CLC202	Dr Sukanya Pandey

Table 2: Michaelmas Term Timetable for Theory of Computation.

2 Models of Computation

This section is taught by Dr Barnaby Martin.

2.1 Finite-State Automata and Regular Languages

2.2 Büchi Automata and Omega-Regular Languages

2.3 Linear-Time Logic and Translating Into Büchi Automata

3 Algorithms and Complexity

This section is taught by Dr Sukanya Pandey. The recommended reading for this topic is:

- Introduction to Algorithms (Cormen T.H., et al., 2009).

3.1 Asymptotic Complexity and Notation

3.2 Greedy Algorithms

3.3 Divide and Conquer Algorithms

3.4 Dynamic Programming Algorithms

3.5 Machinings in Graphs

Part I

Appendix

4 Bibliography

4.1 Module Professors

Module Lead: Doctor Barnaby Martin

- Focus: Complexity theory, finite model theory, and the links between logic and complexity.
- Email: barnaby.d.martin@durham.ac.uk
- Office Hours: Flexible, contact to arrange.

Lecturer: Doctor Sukanya Pandey

- Focus: Graph theory.
- Email: sukanya.pandey@durham.ac.uk
- Office Hours: Flexible, contact to arrange.

Lecturer: Doctor Stefan Dantchev

- Focus: Combinatorics.
- Email: s.s.dantchev@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor George Mertzios

- Focus: Algorithms and networks.
- Email: george.mertzios@durham.ac.uk
- Office Hours: **To be confirmed.**

4.2 References

4.3 List of Figures

4.4 Recommended Reading

4.4.1 Algorithms and Complexity

Cormen, T.H., Leiserson, C.E., Rivest, R.L. and Stein, C. (2009). Introduction to Algorithms. 3rd edn. MIT Press.